

Ankit Kumar

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PROFESSIONAL SUMMARY

Senior Software Engineer from **IIT Delhi** with 3+ years of experience building and deploying production AI/ML and LLM systems spanning predictive modeling, RAG, model inference, fine-tuning, evaluation, and MLOps. Experienced in **Python, PyTorch, TensorFlow, scikit-learn, LangChain, vLLM, Kubernetes, and GCP**, with a focus on model quality, scalable inference, reliability, and production deployment.

WORK EXPERIENCE

Wells Fargo, Hyderabad

July 2023 - Present

• Senior Software Engineer | Machine Learning & LLM Systems

- * Deployed and optimized **LLM** inference services on **Run:ai** using **vLLM**, tuning serving configurations and resource allocation to improve throughput and latency targets, meeting **sub-10-second** response time requirements.
- * Fine-tuned open-source **LLMs** including **gpt-oss & Gemma 3** using **LoRA & QLoRA**, improving task-specific model performance by **10–15%** while reducing training memory requirements by up to **70%**.
- * Built an automated **MLOps** deployment pipeline on **OpenShift AI** using **Argo CD**, validating target cluster and hardware requirements before deploying inference workloads triggered by Git-based configuration changes.
- * Developed a proactive failure-prediction pipeline using a 15-minute horizon, applying **PCA**-based dimensionality reduction and evaluating **LSTM** and **XGBoost** models that achieved **~90%** accuracy in predicting system alerts.
- * Developed testbeds for **generation and guardrail** inference workloads to benchmark model-serving latency, reliability and validate production readiness against **sub-10-second** response-time requirements.

• Software Development Engineer | Applied AI & LLM Platforms

- * Developed an **LLM-powered** back-book validation system to verify client loan records against master databases, reducing validation time and operational costs by **~30%**; the solution secured **\$4.8M** in organizational funding.
- * Developed an **LLM evaluation** framework using **Giskard and DeepEval** with a custom LLM-as-a-judge pipeline, evaluating response relevancy, groundedness, summarization quality, and model performance across candidate LLMs.
- * Built an end-to-end **RAG pipeline** integrating Document AI, embeddings, BigQuery vector storage, vector indexing, semantic search, reranking, grounding, and Gemini APIs to deliver reliable context-aware responses.
- * Implemented end-to-end **observability and tracing** for LLM and multi-agent workflows using **Arize**, enabling production monitoring, debugging, and root-cause analysis across model execution pipelines.

EDUCATION

Indian Institute of Technology, Delhi

2019 - 2023

Bachelor of Technology in Civil Engineering

CGPA: 8.484/10

TECHNICAL SKILLS

Programming Languages: C++, Python, SQL

Frameworks/Libraries: LangChain, LangGraph, LangSmith, LlamaIndex, vLLM, AutoGen, DeepEval, Giskard, FastAPI, scikit-learn, XGBoost, TensorFlow, PyTorch, Matplotlib, OpenAI Python SDK, Google Cloud Client Libraries

Infrastructure/Tools: GCP, Kubernetes, Run:ai, OpenShift AI, Terraform, Argo CD, PostgreSQL, Arize, Redis

Machine Learning & GenAI: RAG, LLM Fine-Tuning, LoRA, QLoRA, LLM Evaluation, LLM-as-a-judge, Vector Search, Embeddings, Semantic Search, Reranking, Grounding, Multi-Agent Systems, Model Serving, Inference Optimization, LLM Observability

ACHIEVEMENTS

- Secured **All India Rank 2,500** in JEE Advanced 2019 among 161,000+ candidates nationwide.
- Received the **Top 7** Percentile Academic Award for two consecutive semesters at IIT Delhi.
- Promoted to **Senior Software Engineer** in recognition of high-impact contributions to production AI/ML initiatives.